

Chao TANG

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Education

University of Macau

Bachelor of Science, Computer Science

Sep 2023 – Present

Relevant Courses: Natural Language Processing; Information Retrieval and Web Search; Linear Algebra I; Probability and Statistics; Discrete Structures; Cloud Computing and Big Data Systems.

Research Interests

- LLM-based agents: self-referential and self-evolving agent frameworks, and RL for agentic tasks.
- AI Scientist: autonomous AI-scientist systems coupled with scalable real-world feedback (e.g., automated wet-lab experimentation).

Research Experience

LLM-Guided Protein Variant Screening under Limited Query Budgets

Sep 2026 – Present

Final-Year Project, Supervised by Prof. Meng QU

University of Macau

- Built an offline protein-variant screening framework on FLIP2 TrpB using Ridge prediction, bootstrap uncertainty estimation, and three batch acquisition strategies under a 48-query budget.
- Implemented rule-based and LLM acquisition controllers; conducted an input-information ablation study with matched baselines comprising 228 trajectories and 456 LLM decisions.
- Compared batch protocols across 19 paired seeds, observing 30.26 versus 25.16 mean top-decile hits for LLM-controlled batches of 4 versus 16 under the same 48-query budget.

CARE: Context-Aware Ranking Evolution with Executable Scoring Programs for Budgeted Reaction Optimization

Research Assistant, Supervised by Dr. Tianyu SHI & Dr. Peiyu ZHANG

McGill University & XtalPi Inc.

Co-first author | arXiv:2606.14581 | Under review at AAI 2027

May 2026 – Present

- Designed an intervention gate that compares candidates from LLM-generated scoring programs with a numerical optimizer's reference choice to limit unreliable or detrimental interventions, achieving 2.9%–7.8% reductions in mean normalized regret across three reaction-optimization tasks in gate ablations.
- Conducted LLM-only baseline comparisons and component ablations to examine the benefits and limitations of language-model intervention in budgeted reaction optimization.
- Identified additional reaction datasets to support evaluation of CARE across diverse reaction settings.
- Developed robust evaluation processes and achieved the lowest mean normalized regret on each of eight reaction-optimization tasks among the evaluated methods, using five initial observations and ten adaptive experiment selections per campaign in offline replay.

Agent-guided Microfluidic Automation for Biochemical Assays

May 2026 – Present

Visiting Student, Supervised by Prof. Haisong LIN at Laboratory for Biochips & Biosensors

Westlake University

- Designed and implemented the overall LLM-agent system for adapting biochemical assay-kit protocols to an oil–water interfacial microfluidic platform, connecting protocol interpretation with experiment planning toward automated biochemical detection.
- Implemented generation of microfluidic chamber-layout designs and laboratory standard operating procedures (SOPs) from assay protocols, translating kit instructions into platform-specific experimental designs and procedures for laboratory personnel.
- Developed software drivers for PCB-mounted coils to actuate magnets for manipulating magnetic beads or ferrofluids; established the software-to-hardware connection for the microfluidic platform for magnetic motion demonstration on physical PCB hardware.
- Contributed to devising the agent evaluation plan by developing figures presenting the system overview, agent workflow, and performance characterization, supporting ongoing wet-lab assay demonstrations.

Agent-Based Graphic Design and Editable SVG Generation

Feb 2025 – Present

Lab Member, Supervised by Prof. Fai WONG at NLP2CT

University of Macau

- Designed and implemented an agent-based graphic-design system to generate editable SVG layouts and visual content from natural-language briefs and style preferences.
- Built a staged workflow for element identification, occlusion detection, hidden-content regeneration, and layer-order determination to address overlapping design elements, recovering obscured content and separating elements for independent editing.
- Developed agent-guided segmentation with the Segment Anything Model (SAM), using object-specific language prompts to address incorrect or extraneous masks during visual-element extraction.
- Combined optical character recognition (OCR) with background reconstruction to replace rasterized lettering with editable text elements, restoring the image content beneath the original text before rebuilding the text layer.
- Implemented a ReAct agent to inspect element placement and visual composition, and iteratively refine layouts; assembled extracted visual assets and reconstructed text in the SVG's XML structure to preserve element-level editability.
- Selected 200 examples from LICA (Layered Image Composition Annotations), a layered graphic-design dataset, for text-to-design evaluation.
- Currently building an LLM-as-a-judge assessment of instruction adherence and visual aesthetics, and full-set evaluation.

Leadership

President of the Artificial Intelligence and Data Science Society, University of Macau Students' Union Aug 2025 – Present

Student leader of MLC College, University of Macau

Aug 2024 – Jun 2025

President of the Photography Society, University of Macau Students' Union

Aug 2024 – Jun 2025

Skills

Programming & Tools: Python, Hugging Face Transformers, Linux, Git, LaTeX.